

Camera Sensor 适配流程指南

参考文档:

《Camera Sensor 适配指南》

《Sensor 调试指南》

《MIPI 使用指南》

《SYS_CONFIG 配置指南》

《pq 环境搭建.docx 》

第 1 章 适配前准备

在动手写代码前, 必须从 sensor 厂获取以下资料 (参考《Camera Sensor 适配指南》1.1 节):

- Sensor Datasheet 如: IMX678-AAQR1-C_E_Datasheet_E21Y08B34.pdf
- Sensor Design Reference Manual 如: IMX678_SoftwareReferenceManual_E_Rev5.0.ptf
- Sensor Register Setting 如: IMX678_Register_Map_Standard_Register_Setting.xlsx
- Gain Table 如: OS04J10 Gain Table.xlsx
- 初始化序列, 申请格式通常如下:

Sensor 型号: OS04J10

主控: HI3519AV200

分辨率: 2560*1440

帧率: 30FPS

输入时钟: 24MHz

输出模式: Linear

接口类型: MIPI 4Lane

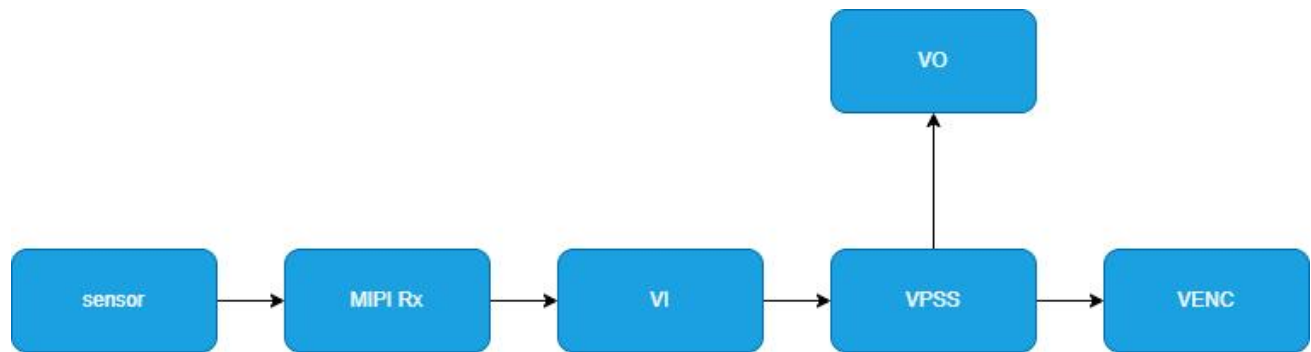
数据格式: RAW 12bit

master 模式

[注意] 这些文档要备份在: \\192.168.100.222\public\器件规格书\Sensor

第 2 章 Camera 数据通路

一颗 Camera Sensor 从光信号到屏幕显示/录像存储, 数据要经过下图所示的链路:



Camera Sensor 数据通路

其中 Sensor 通过两条独立的链路与 SOC 交互:

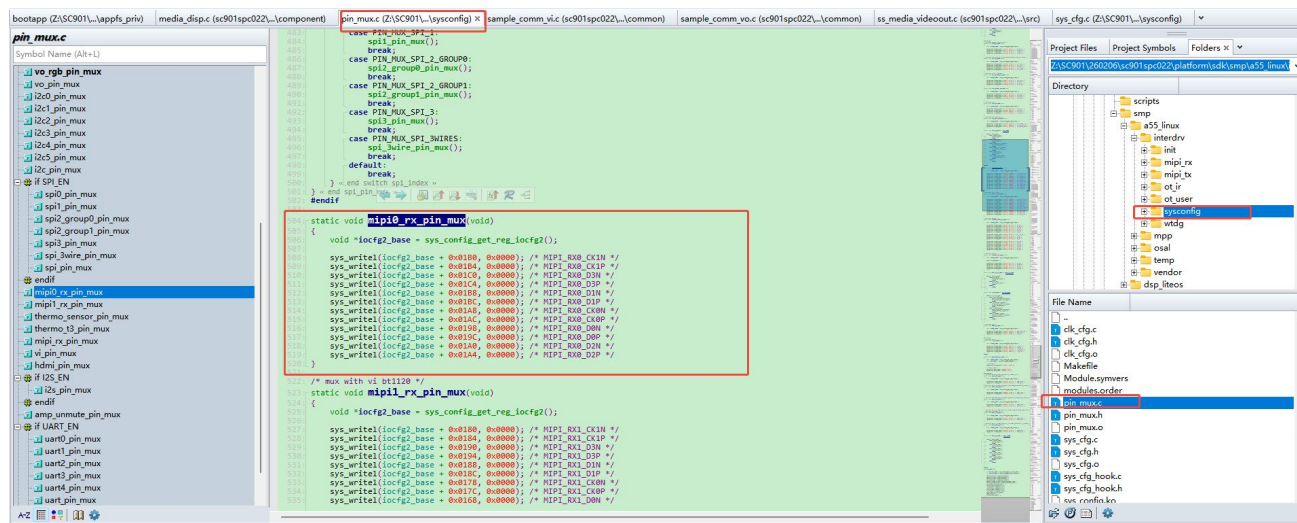
- I2C 控制链路: SOC 通过 I2C 写 sensor 寄存器, 完成上电初始化、AE 增益、曝光时间下发
- MIPI 数据链路: Sensor 通过 D-PHY 差分信号 (CSI-2 协议) 把图像数据高速送到 SOC MIPI Rx, 经 VI 进入媒体处理链路

第3章 sensor 出图

3.1

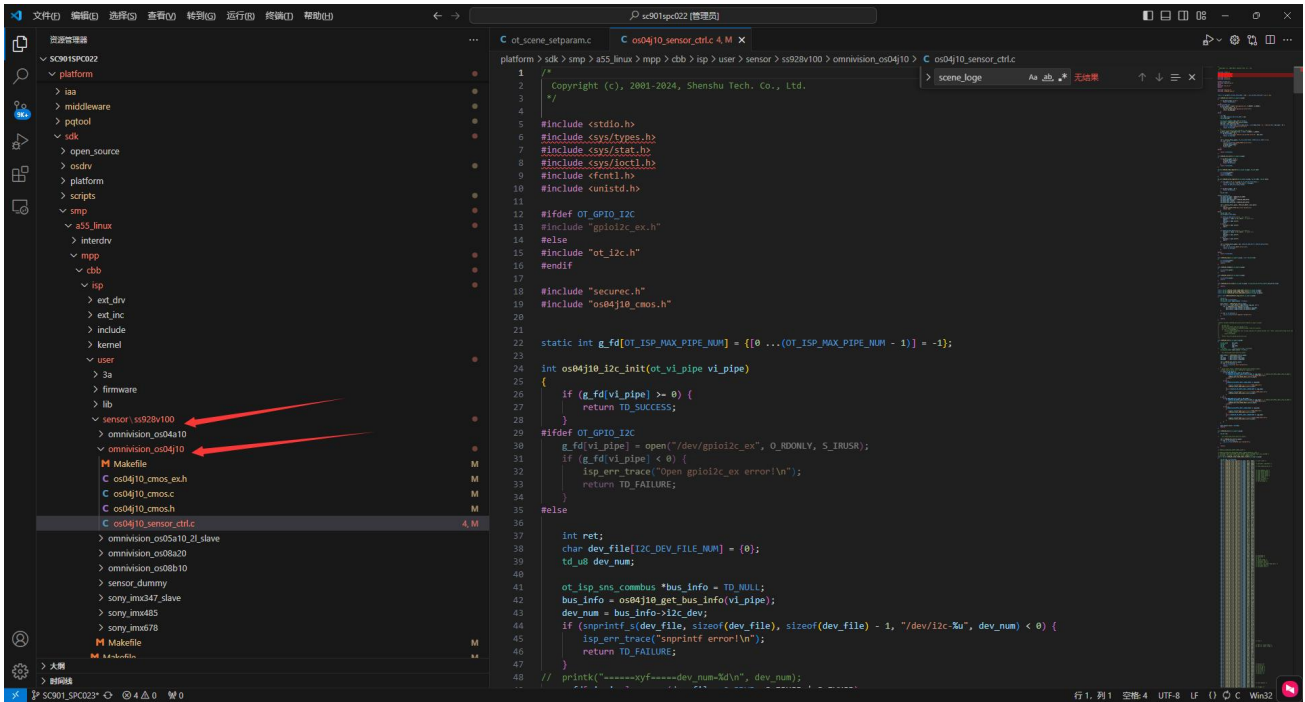
根据项目的 IO list 和《SYS_CONFIG 配置指南》，在 pin_mux.c 中确定 sensor 的的时钟、复位、MIPI、I2C 的 IO 复用正确。

	A	B	C	D	E	F
7		I2C3_SDA	SENSOR I2C数据。	I2C3_SDA/GPIO16_6	SENSOR I2C数据接口	
8		MIPI_RX0_D0N		MIPI_RX0_D0N/GPIO14_3		
9		MIPI_RX0_D0P		MIPI_RX0_D0P/GPIO14_4		
10	路IMX678 SENS	MIPI_RX0_D1N		MIPI_RX0_D1N/GPIO15_3		
11		MIPI_RX0_D1P	4lane MIPI信号数据	MIPI_RX0_D1P/GPIO15_4		
12		MIPI_RX0_D2N		MIPI_RX0_D2N/GPIO14_5		
13		MIPI_RX0_D2P		MIPI_RX0_D2P/GPIO14_6		
14		MIPI_RX0_D3N		MIPI_RX0_D3N/GPIO15_5		
15		MIPI_RX0_D3P		MIPI_RX0_D3P/GPIO15_6		
16		MIPI_RX0_CK0N		MIPI_RX0_CK0N/GPIO14_7		
17		MIPI_RX0_CK0P	MIPI信号时钟	MIPI_RX0_CK0P/GPIO15_0		
18		SENSOR1_CLK	Image Sensor1工作时钟。	SENSOR1_CLK/GPIO12_1		
19		SENSOR1_RSTN	Sensor复位信号。	SENSOR1_RSTN/GPIO12_0		
20		I2C2_SCL	SENSOR I2C时钟信号。	I2C2_SCL/GPIO16_3		
21		I2C2_SDA	SENSOR I2C时钟信号。	I2C2_SDA/GPIO16_4		
22		MIPI_RX1_D0N		MIPI_RX1_D0N/GPIO12_7		
23		MIPI_RX1_D0P		MIPI_RX1_D0P/GPIO13_0		
24	内录OS04J10	MIPI_RX1_D1N		MIPI_RX1_D1N/GPIO13_7		
25	Sensor	MIPI_RX1_D1P		MIPI_RX1_D1P/GPIO14_0		
26		MIPI_RX1_D2N	2lane MIPI信号数据	MIPI_RX1_D2N/GPIO13_1		
27		MIPI_RX1_D2P		MIPI_RX1_D2P/GPIO13_2		
28		MIPI_RX1_D3N		MIPI_RX1_D3N/GPIO14_1		
29		MIPI_RX1_D3P		MIPI_RX1_D3P/GPIO14_2		
30		MIPI_RX1_CK0N		MIPI_RX1_CK0N/GPIO13_3		
31		MIPI_RX1_CK0P		MIPI_RX1_CK0P/GPIO13_4		

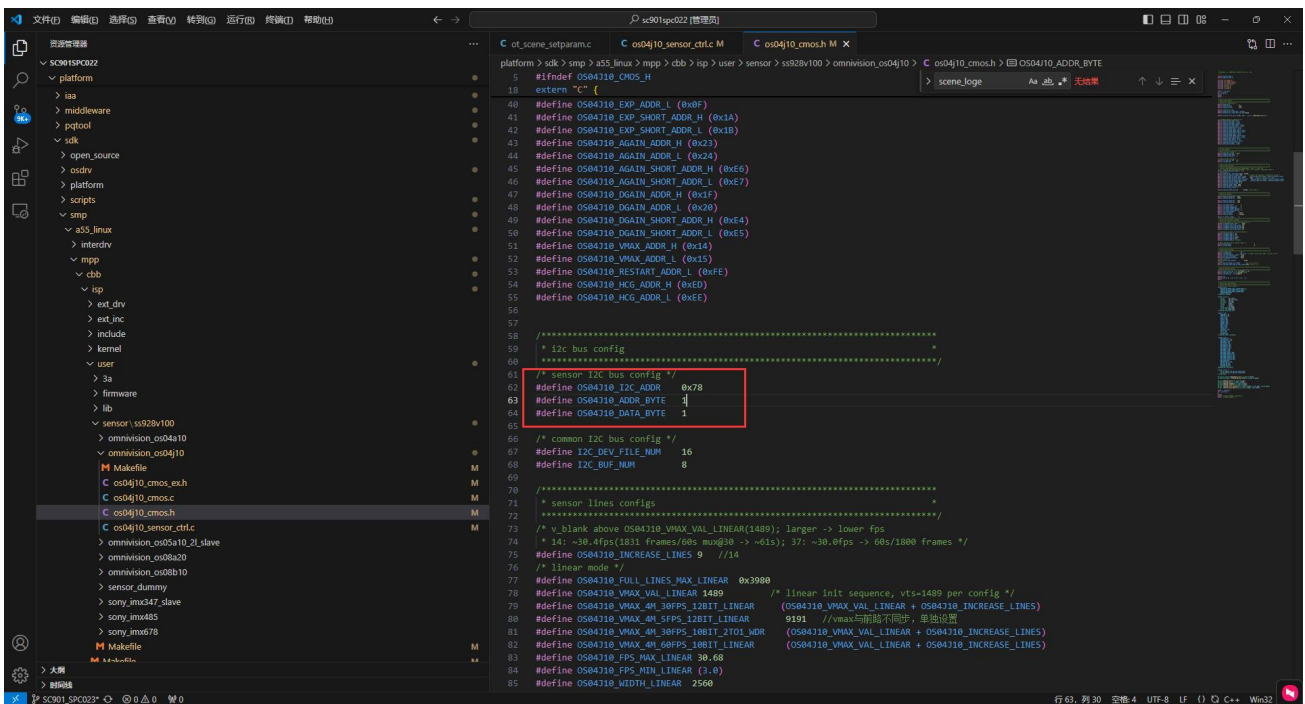


3.2

在 platform/sdk/smp/a55_linux/mpp/cbb/isp/user/sensor/ss928v100/目录下，模仿其它已经调通的 sensor 新建一个驱动目录，通常是复制一个与你要调的 sensor 相似的型号的驱动过来，然后再进行修改。比如我要调试 os04j10 这个 sensor 的 linear 模式，就要新建一个如下图的驱动目录



3.3 根据 sensor 规格书修改 i2c 的设备地址和寄存器的长度



检查 sensor 的 i2c 地址

检查地址长度,

例如 sensor 的某个寄存器的地址是 0x36D8 (像 imx678)

```
#define OS04J10_ADDR_BYTE 2
```

```
#define OS04J10_DATA_BYTE 1
```

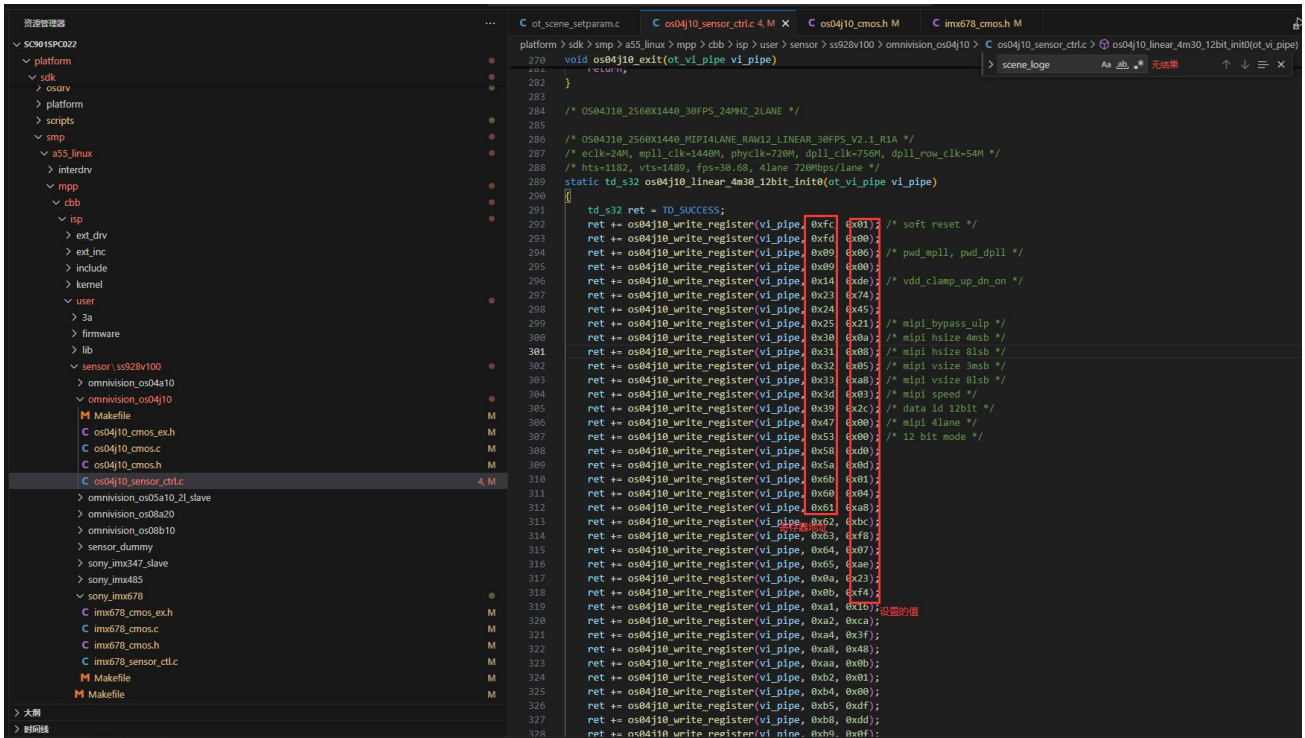
再例如 sensor 的某个寄存器地址是 0x5e (像 os04j10)

```
#define OS04J10_ADDR_BYTE 1 这里需要注意
```

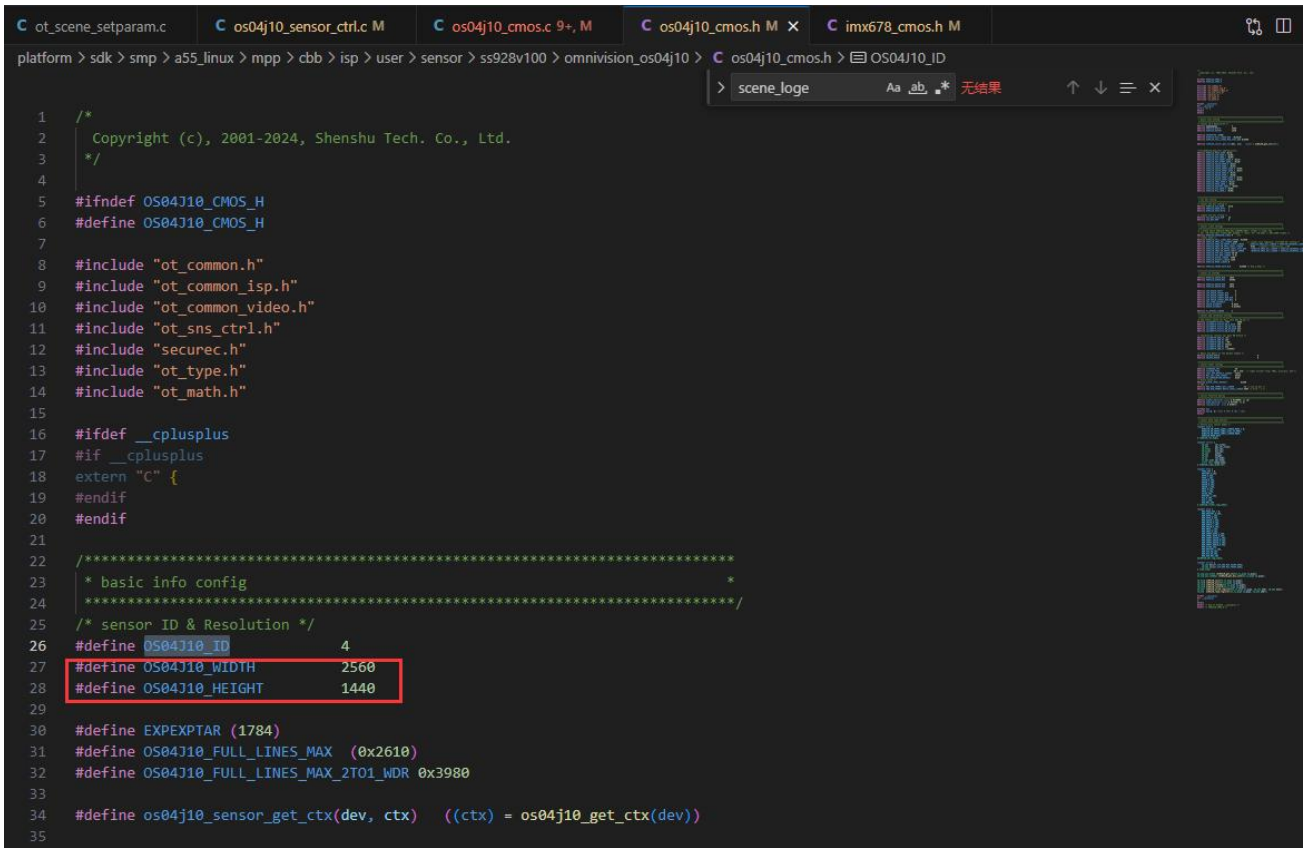
```
#define OS04J10_DATA_BYTE 1
```

3.4 修改 sensor 初始化序列。

```
35 78 fd 00
36 78 09 06 ;pwd_mpll, pwd_dpll
37 78 09 00 ;
38 78 14 de ;vdd_clamp_up_dn_on
39 78 17 54 ;dpll_nc
40 78 1d 78 ;mpll_nc
41 78 23 74 ;
42 78 24 45 ;
43 78 25 21 ;mipi_bypass_ulp
44 78 30 0a ;mipi_hsize 4msb
45 78 31 00 ;mipi_hsize 8lsb
46 78 32 05 ;mipi_vsize 3msb
47 78 33 a0 ;mipi_vsize 8lsb
48 78 3d 01 ;mipi_speed
49 78 47 00 ;mipi_4lane
50 78 58 d0
51 78 5a 0d
52 78 6b 01
53 78 60 04
54 78 61 a8
55 78 62 bc
56 78 63 f8
57 78 64 07
58 78 65 ae
59 设备地址 78 0a 23
60 78 0b f4
61 78 a1 16
62 寄存器地址 78 a2 ca
63 78 a4 3f
64 78 a8 41
65 78 aa 00 设置的值
66 78 b2 01
67 78 b4 00
68 78 b5 df
69 78 b8 dd
70 78 b9 0f
71
```



3.5 修改宽高信息，帧率信息



The screenshot shows a code editor with several tabs at the top: `ot_scene_setparam.c`, `os04j10_sensor_ctrl.c M`, `os04j10_cmos.c 9+, M`, `os04j10_cmos.h M X`, and `imx678_cmos.h M`. The active tab is `os04j10_cmos.h`, which displays the following C code:

```
1  /*
2   Copyright (c), 2001-2024, Shenshu Tech. Co., Ltd.
3   */
4
5  #ifndef OS04J10_CMOS_H
6  #define OS04J10_CMOS_H
7
8  #include "ot_common.h"
9  #include "ot_common_isp.h"
10 #include "ot_common_video.h"
11 #include "ot_sns_ctrl.h"
12 #include "securec.h"
13 #include "ot_type.h"
14 #include "ot_math.h"
15
16 #ifdef __cplusplus
17 #if __cplusplus
18 extern "C" {
19 #endif
20 #endif
21
22 /******
23  * basic info config
24  *****/
25 /* sensor ID & Resolution */
26 #define OS04J10_ID 4
27 #define OS04J10_WIDTH 2560
28 #define OS04J10_HEIGHT 1440
29
30 #define EXPEXPTAR (1784)
31 #define OS04J10_FULL_LINES_MAX (0x2610)
32 #define OS04J10_FULL_LINES_MAX_2T01_WDR 0x3980
33
34 #define os04j10_sensor_get_ctx(dev, ctx) ((ctx) = os04j10_get_ctx(dev))
35
```

The values `2560` and `1440` for `OS04J10_WIDTH` and `OS04J10_HEIGHT` are highlighted with a red box. On the right side of the editor, a search bar contains the text `scene_loge` and the results show `无结果` (No results).

```
C ot_scene_setparam.c C os04j10_sensor_ctrl.c M C os04j10_cmos.c 9+, M X C os04j10_cmos.h M C imx678_cmos.h M
platform > sdk > smp > a55_linux > mpp > cbb > isp > user > sensor > ss928v100 > omnivision_os04j10 > C os04j10_cmos.c > g_os04j10_mode_tbl
37
38 static td_u16 g_init_wb_gain[OT_ISP_MAX_PIPE_NUM][OT_ISP_RGB_CHN_NUM] = {0};
39 static td_u16 g_sample_r_gain[OT_ISP_MAX_PIPE_NUM] = {0};
40 static td_u16 g_sample_b_gain[OT_ISP_MAX_PIPE_NUM] = {0};
41 static td_bool g_quick_start_en[OT_ISP_MAX_PIPE_NUM] = {TD_FALSE};
42
43 static td_bool g_ae_route_ex_valid[OT_ISP_MAX_PIPE_NUM] = {0};
44 static ot_isp_ae_route g_init_ae_route[OT_ISP_MAX_PIPE_NUM] = {{0}};
45 static ot_isp_ae_route_ex g_init_ae_route_ex[OT_ISP_MAX_PIPE_NUM] = {{0}};
46 static ot_isp_ae_route g_init_ae_route_sf[OT_ISP_MAX_PIPE_NUM] = {{0}};
47 static ot_isp_ae_route_ex g_init_ae_route_sf_ex[OT_ISP_MAX_PIPE_NUM] = {{0}};
48
49 static ot_isp_sns_commbus g_bus_info[OT_ISP_MAX_PIPE_NUM] = {
50     [0] = { .i2c_dev = 0 },
51     [1 ... OT_ISP_MAX_PIPE_NUM - 1] = { .i2c_dev = -1 }
52 };
53
54 static ot_isp_sns_state *g_os04j10_sns_state[OT_ISP_MAX_PIPE_NUM] = {TD_NULL};
55
56 static td_bool blc_clamp_info[OT_ISP_MAX_PIPE_NUM] = {[0 ... (OT_ISP_MAX_PIPE_NUM - 1)] = TD_TRUE};
57
58 /* key params of your sensor modes */
59 /* linear 模式 min_fps 由 25 降到 OS04J10_FPS_MIN_LINEAR (3.0),
60 * 让延时录像可以把 sensor 物理帧率压到 5fps: cmos_fps_set 会按
61 * vmax = ver_lines * max_fps / fps = 1526 * 30 / 5 = 9156 改写 OS04J10
62 * 的 VMAX 寄存器, sensor 每帧周期变长 6 倍, max_int_time 同步扩大,
63 * 真正降低 sensor/ISP 功耗并把曝光窗口拉长。9156 < OS04J10_FULL_LINES_MAX
64 * (0x2610=9744), 在硬件 VMAX 范围内。 */
65 const os04j10_video_mode_tbl g_os04j10_mode_tbl[OS04J10_MODE_BUTT] = {
66     {OS04J10_VMAX_4M_30FPS_12BIT_LINEAR, OS04J10_FULL_LINES_MAX, 30.68, OS04J10_FPS_MIN_LINEAR,
67      2560, 1440, 0, OT_WDR_MODE_NONE, "OS04J10_4M_30FPS_12BIT_LINEAR_MODE"},
68     {OS04J10_VMAX_4M_30FPS_10BIT_2TO1_WDR, OS04J10_FULL_LINES_MAX_2TO1_WDR, 30.0, 25.0,
69      2560, 1440, 1, OT_WDR_MODE_2TO1_LINE, "OS04J10_4M_30FPS_10BIT_2TO1_VC_MODE"},
70     {OS04J10_VMAX_4M_60FPS_10BIT_LINEAR, OS04J10_FULL_LINES_MAX, 60, 5,
71      2560, 1440, 2, OT_WDR_MODE_NONE, "OS04J10_4M_60FPS_10BIT_LINEAR_MODE"},
72 };
73
74
```

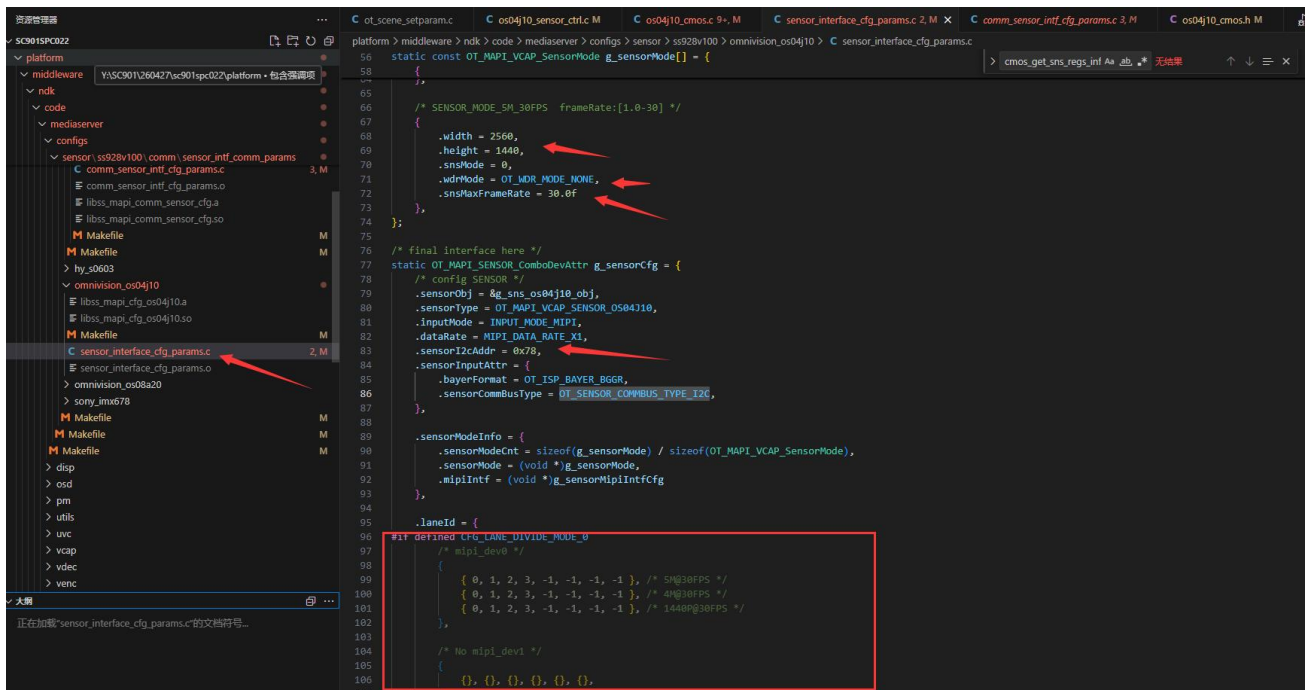
3.6 修改 NDK

增加新 Sensor 的宏定义, 在 ndk/code/include/ot_mapi_vcap_define.h 中添加对应 sensor 的宏定义

```
xyf@SS928:~/share/SC901/260206/sc901spc022$ git diff platform/middleware/ndk/
diff --git a/platform/middleware/ndk/code/include/ot_mapi_vcap_define.h b/platform/middleware/ndk/code/include/ot_mapi_vcap_define.h
index d004f7b..931a0b8 100755
--- a/platform/middleware/ndk/code/include/ot_mapi_vcap_define.h
+++ b/platform/middleware/ndk/code/include/ot_mapi_vcap_define.h
@@ -96,6 +96,7 @@ typedef enum {
     OT_MAPI_VCAP_SENSOR_IMX485,
     OT_MAPI_VCAP_SENSOR_OS08A20,
     OT_MAPI_VCAP_SENSOR_IMX678,
+    OT_MAPI_VCAP_SENSOR_OS04J10,
     OT_MAPI_VCAP_SENSOR_BUTT
 } OT_MAPI_VCAP_SensorType;
```

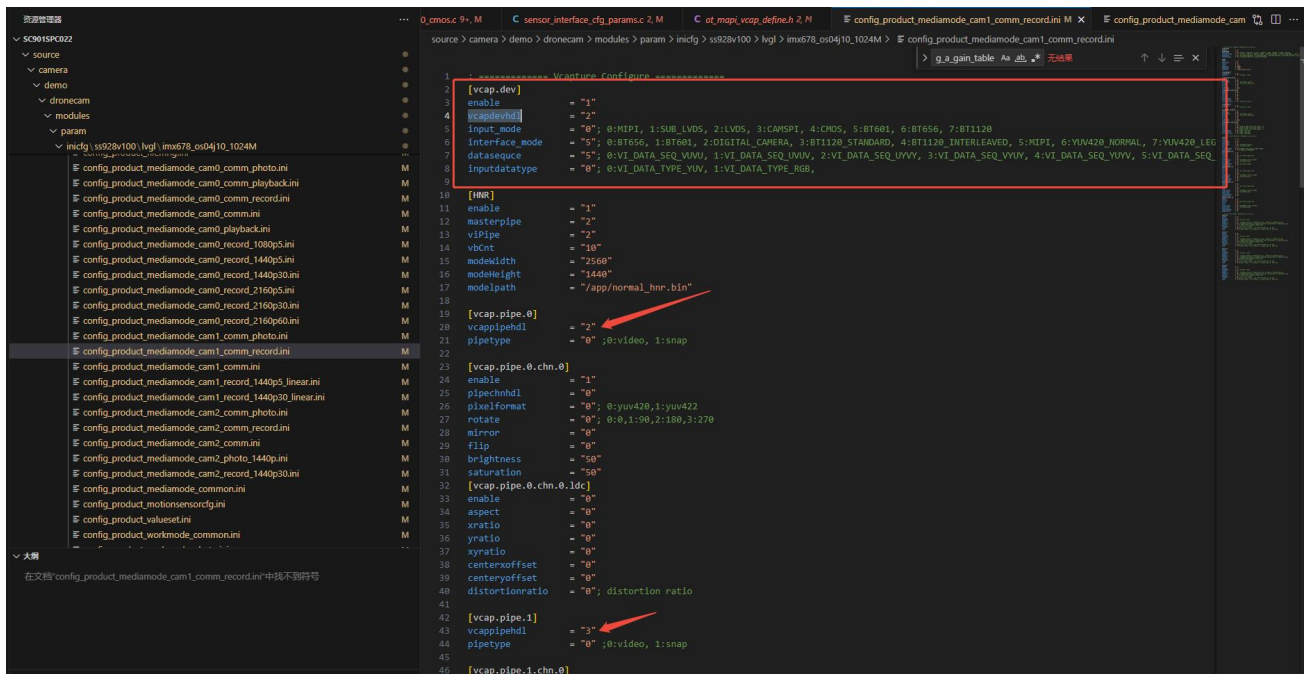
在 ndk/code/mediaserver/configs/sensor/\$(chip)目录下新建一个该 Sensor 的文件夹, 文件夹下新增 sensor_interface_cfg_params.c 和 Makefile 两个文件

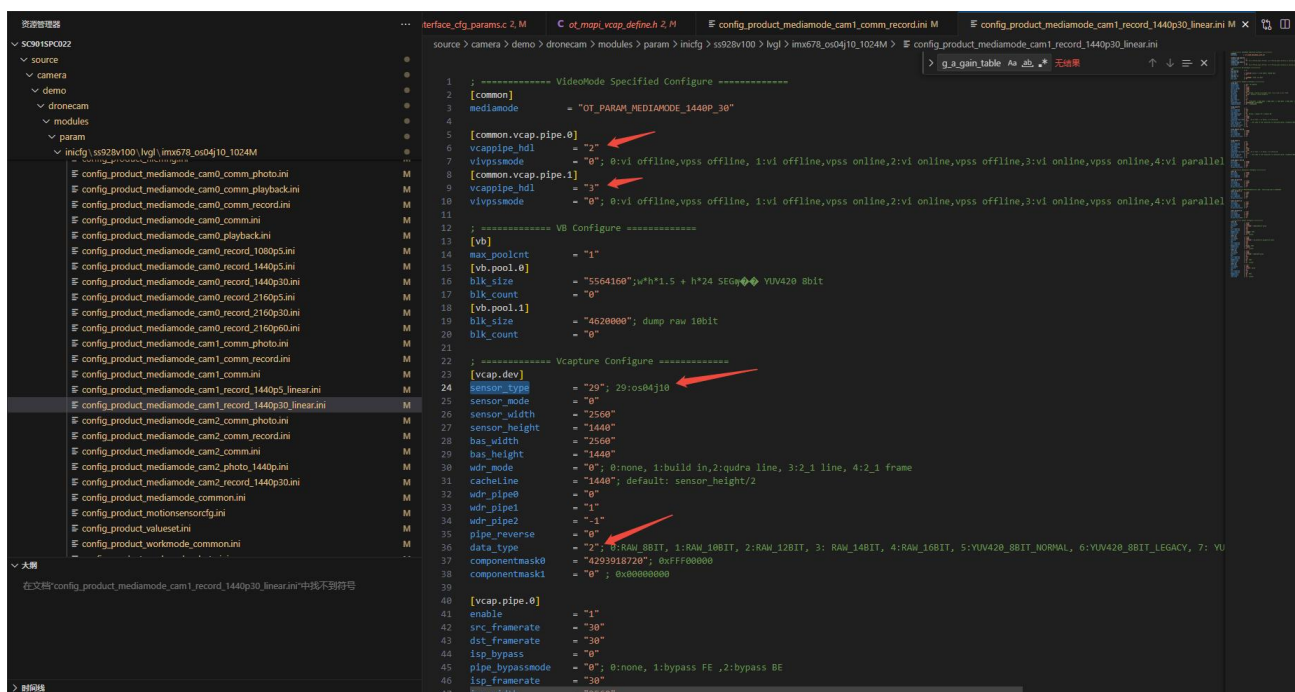
根据《MIPI 使用指南》配置好 sensor_interface_cfg_params.c 这个文件



3.7 修改应用层 ini

主要配置对应 sensor 的 config_product_mediamode_camx_comm_record.ini 和 config_product_mediamode_camx_record_1440p30_linear.ini，把绑定通路和数据格式配置好





3.8 编译开关

在 cmake/build/base.mak, cmake/build/kconfig.mak, cmake/menuconfig/Kconfig.board 分别添加如下编译选项

```

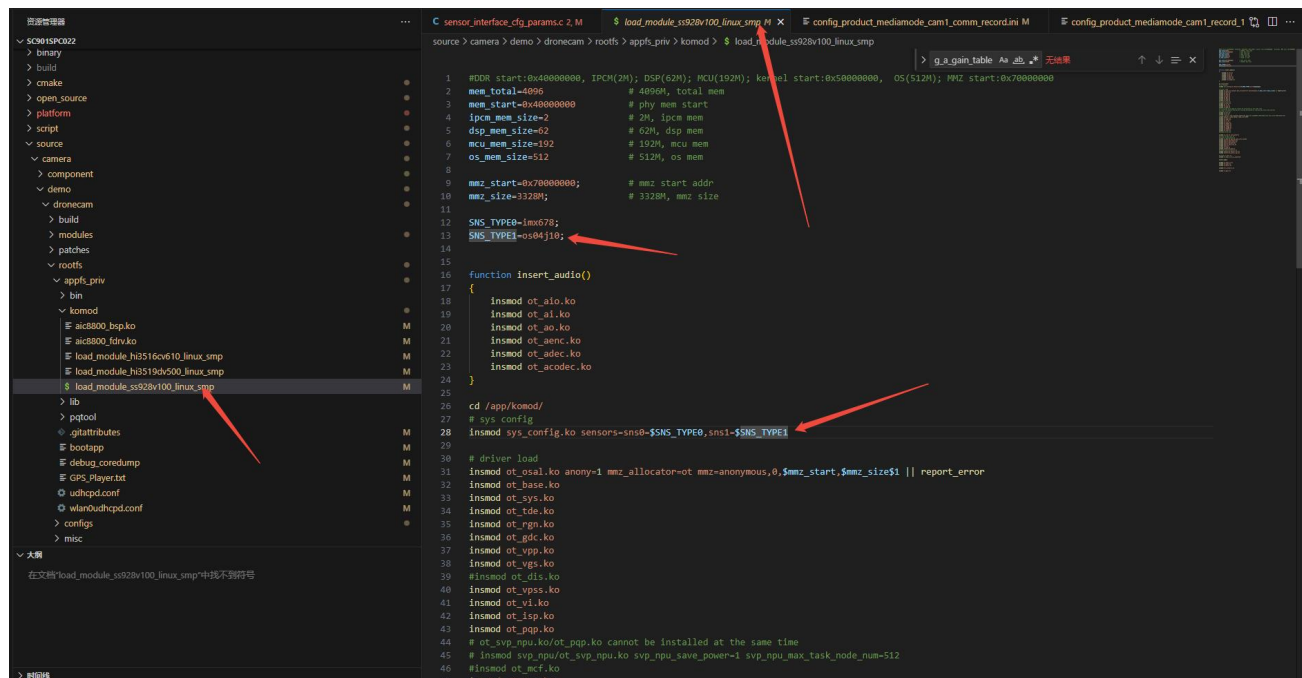
xyf@SS928:~/share/SC901/260206/sc901spc022$ git diff cmake/
build/
menuconfig/
xyf@SS928:~/share/SC901/260206/sc901spc022$ git diff cmake/
diff --git a/cmake/build/base.mak b/cmake/build/base.mak
index af3204b..30af56a 100755
--- a/cmake/build/base.mak
+++ b/cmake/build/base.mak
@@ -189,6 +189,10 @@ else
NDK_COMPILE_SENSOR_LIST +=
endif

+ifeq ($(CONFIG_SNS2_OS04J10), y)
+NDK_COMPILE_SENSOR_LIST += OS04J10
+endif
+
NDK_COMPILE_OPT += CFG_SENSOR_LIST="$(NDK_COMPILE_SENSOR_LIST)"

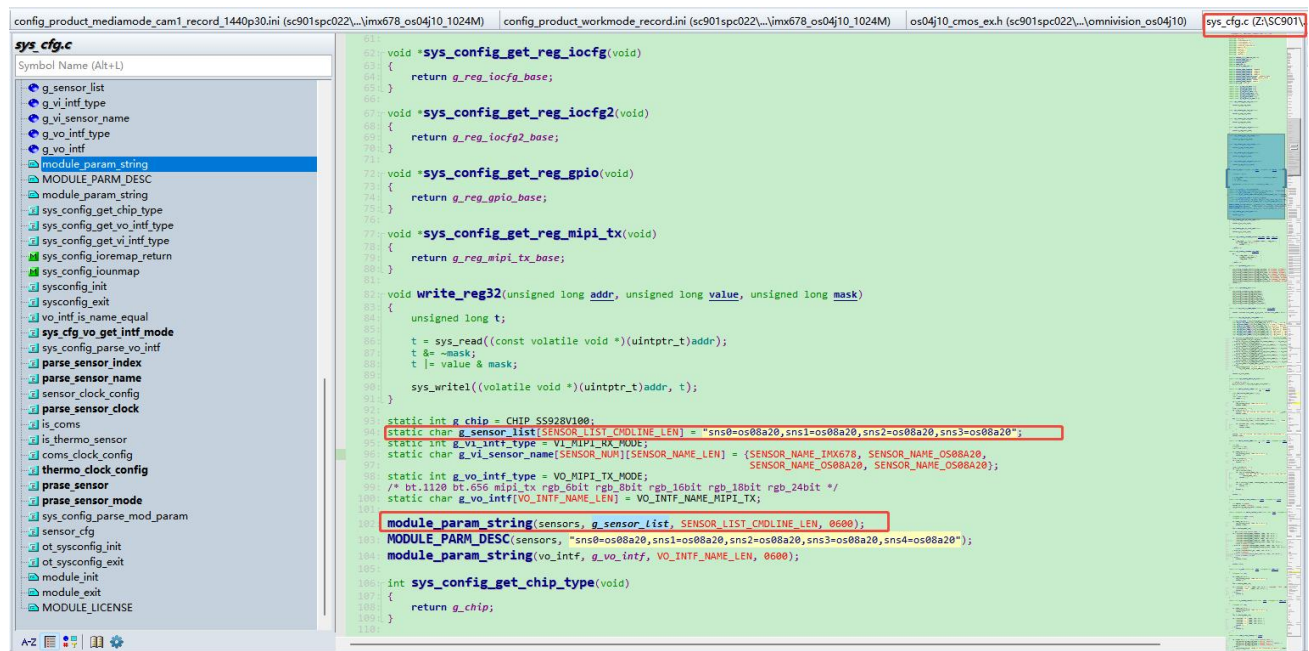
# Compile Option: serdes
diff --git a/cmake/build/kconfig.mak b/cmake/build/kconfig.mak
index cf82ee3..82558a0 100755
--- a/cmake/build/kconfig.mak
+++ b/cmake/build/kconfig.mak
@@ -261,6 +261,9 @@ endif
ifeq ($(CONFIG_SNS1_IMX477), y)
CFG_SENSOR_TYPE1 := imx477
KCONFIG_CFLAGS += -DCONFIG_SNS1_IMX477
+else ifeq ($(CONFIG_SNS2_OS04J10), y)
+CFG_SENSOR_TYPE1 := os04j10
+KCONFIG_CFLAGS += -DCONFIG_SNS2_OS04J10
else ifeq ($(CONFIG_SNS1_IMX307), y)
CFG_SENSOR_TYPE1 := imx307
KCONFIG_CFLAGS += -DCONFIG_SNS1_IMX307
diff --git a/cmake/menuconfig/Kconfig.board b/cmake/menuconfig/Kconfig.board
index 68b872c..63c6ea8 100755
--- a/cmake/menuconfig/Kconfig.board
+++ b/cmake/menuconfig/Kconfig.board
@@ -183,6 +183,9 @@ menu "Sensor Configure"
depends on (CJ103V300_LANE_DIVIDE_MODE_6 || SS927V100)
config SNS2_IMX477
bool "sony_imx477"
+
+config SNS2_OS04J10
+bool "ov_0s04j10"
+depends on (CJ102V500 || SS927V100 || CJ102V700) && DRONECAM
+
config SNS2_NONE
bool "UNUSED"
endchoice

```

同时要修改一下启动脚本 `load_module_ss928v100_linux_smp`，传入正确的 sensor 参数

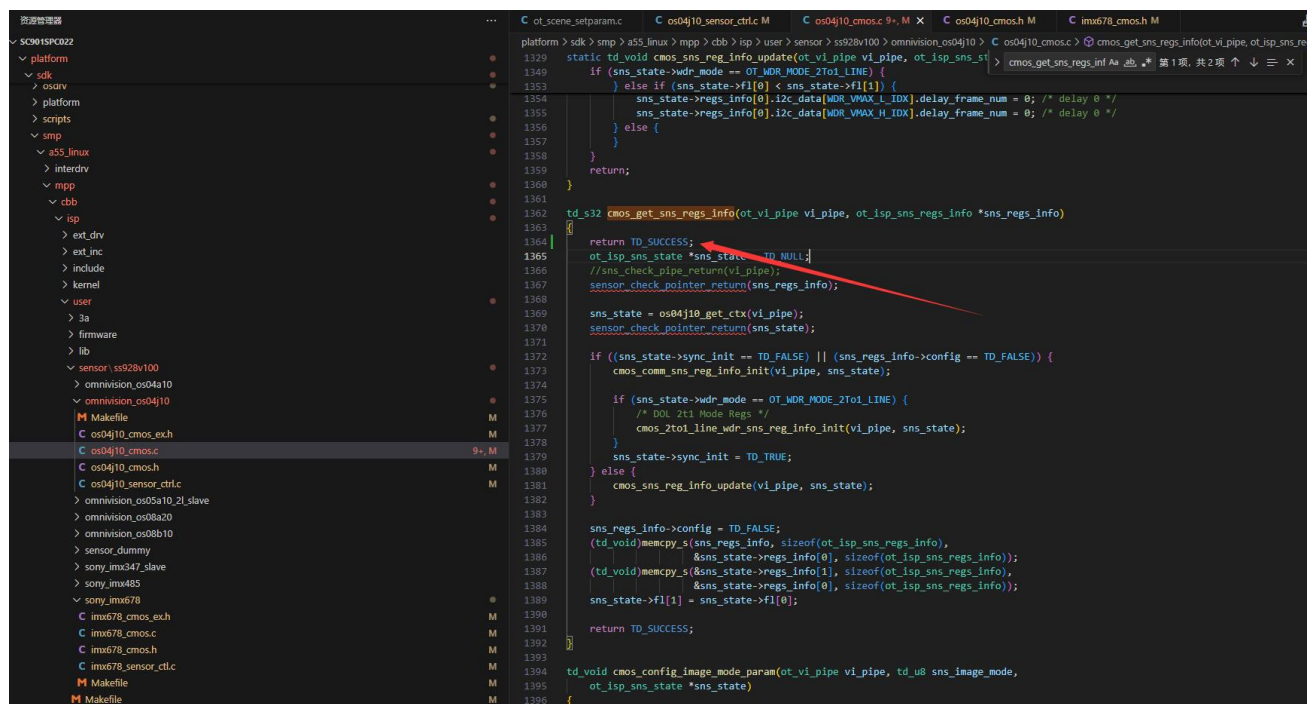


传入的参数会在 sys_cfg.c 这里解析，调试的时候可以在这里加打印确认



3.9 验证 sensor 出图

验证 sensor 出图时，可以认为仅跑初始化序列，可以直接在 cm0s.c 文件中 cm0s_get_sns_regs_info() 函数入口处，直接 return TD_SUCCESS。相当于不使用 isp 中的 ae 参数，也就是使用初始化序列中的默认曝光、增益、帧率配置。到第四章配置 AE 参数时要把这个 return TD_SUCCESS 放开。



验证 sensor 出图时，可以 cat /proc/umap/mipi_rx，查看是否出宽高信息。

```
- #
- # cat /proc/umap/mipi_rx
[MIPi_RX] Version: [SS928V100V1.0.0.0 B010 Release], Build Time[Aug 14 2026, 18:08:57]

-----mipi lane divide mds-----
mode
1
lane divide
4+4

-----mipi dev attr-----
devno work_mode data_rate data_type wdr_mode img_x img_y img_w img_h
0 0 mipi X1 raw10 vc 0 0 3840 2160
2 2 mipi X1 raw12 none 0 0 2560 1440

-----mipi lane info-----
devno lane_id
0 0, 1, 2, 3, -1, -1, -1, -1
2 4, 5, 6, 7, -1, -1, -1, -1

-----mipi phy data info-----
phy_id lane_id phy_data mipi_data lvds_data freq
0 0, 1, 2, 3 0x00,0x00,0x00,0x97 0x6b,0xd4,0x12,0x2f 0xb0,0x0f,0xb0,0x16 179MHz
1 4, 5, 6, 7 0x42,0x4b,0x03,0x7e 0x2c,0x00,0x0f,0x19 0x0f,0x70,0x40,0x5c 89MHz

-----mipi detect info-----
devno vc width height
0 0 3840 2160
0 1 3840 2160
0 2 0 0
0 3 0 0
2 0 2560 1440
2 1 0 0
2 2 0 0
2 3 0 0

-----phy cil err int info-----
phy_id clk2_tmout clk_tmout lane0_tmout lane1_tmout lane2_tmout lane3_tmout clk2_esc clk_esc lane0_esc lane1_esc lane2_esc lane3_esc
0 0 0 0 0 0 0 0 0 0 0 0
1 0 0 0 0 0 0 0 0 0 0 0

-----mipi error int info1-----
devno ecc2 vc0_crc vc1_crc vc2_crc vc3_crc vc0_ecc_correct vc1_ecc_correct vc2_ecc_correct vc3_ecc_correct
0 0 0 0 0 0 0 0 0 0
2 0 0 0 0 0 0 0 0 0

-----mipi error int info2-----
devno vc0_frmnum vc1_frmnum vc2_frmnum vc3_frmnum vc0_frmmatch vc1_frmmatch vc2_frmmatch vc3_frmmatch
0 0 0 0 0 0 0 0
2 1 0 0 0 0 0 0

-----mipi error int info3-----
devno data_fifo_rderr cmd_fifo_rderr vsync cmd_fifo_wrerr data_fifo_wrerr
0 0 0 0 0 0
2 0 0 0 0 0

-----align error int info-----
devno fifo_fullerr lane0_err lane1_err lane2_err lane3_err lane4_err lane5_err lane6_err lane7_err
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

是否有数据

宽高是否正确

检查 vi 中是否有帧率。

```

-----vi phys chn zoom in window attr-----
pipe_id  chn_id  enable  mode    x      y      width  height  trim_x  trim_y  trim_width  trim_height  valid
0         0       N      ABS     0      0      0      0      0      0      0          0          0      N
1         0       N      ABS     0      0      0      0      0      0      0          0          0      N
2         0       N      ABS     0      0      0      0      0      0      0          0          0      N

-----vi phys chn lut attr-----
pipe_id  chn_id  pre_lut_en  post_lut_en  lut_mode  ratio
0         0       N           N           N      connect  -
1         0       N           N           N      connect  -
2         0       N           N           N      connect  -

-----vi phys chn lut-----
pipe_id  chn_id  name  in_width  in_height  out_width  out_height  cell_size  phys_addr

-----vi phys chn fisheye attr-----
pipe_id  chn_id  enable  mount_mode  rgn_num  bg_color_en  bg_color  lmf_en  hor_offset  ver_offset  trap_coef  fan_strength
0         0       N      DESKTOP    0         N           0x0      N        0           0           0          0
1         0       N      DESKTOP    0         N           0x0      N        0           0           0          0
2         0       N      DESKTOP    0         N           0x0      N        0           0           0          0

-----vi phys chn fisheye rgn attr-----
pipe_id  chn_id  rgn_index  view_mode  in_radius  out_radius  pan  tilt  hor_zoom  ver_zoom  out_x  out_y  out_w  out_h

-----vi phys chn fov correction attr-----
pipe_id  chn_id  enable
0         0       N
1         0       N
2         0       N

-----vi ext chn attr1-----
pipe_id  chn_id  bind_chn  src_type  width  height  pixel_format  dynamic_range  video_format  compress_mode

-----vi ext chn attr2-----
pipe_id  chn_id  depth  src_rate  dst_rate  vb_src  vb_pool  align

-----vi chn low delay attr-----
pipe_id  chn_id  enable  line_cnt  one_buf
0         0       N        0        N
2         0       N        0        N

-----vi chn crop info-----
pipe_id  chn_id  crop_en  coord  x      y      width  height  trim_x  trim_y  trim_width  trim_height
0         0       N      ABS     0      0      0      0      0      0      0          0
2         0       N      ABS     0      0      0      0      0      0      0          0

-----vi chn status-----
pipe_id  chn_id  enable  width  height  receive_cnt  vb_fail_cnt  lost_frame_cnt  frame_rate
0         0       Y    3840   2160      50668        0             0             30
2         0       Y    2560   1440      50659        0             0             30

-----vi chn out frame info-----
pipe_id  chn_id  width  height  compress_mode  low_delay_frame  delay
0         0    3840   2160      seg           N             49565
2         0    2560   1440      none          N             135016

```

第 4 章 驱动 AE 参数配置

完成第 3 章节的内容后，sensor 已经具备了基本的出图能力，接下来需要适配 ae 参数，其中包括增益、曝光、帧率控制三个方面。

阅读sensor datasheet手册 收集以下信息

		说明
i2c_dev	0	连接板端的设备号
i2c_addr	0x	Sensor的i2c设置地址
Sensor_CLK	24M/27M/?	Sensor的时钟频率
VMAX_ADDR_L	0x	帧率寄存器地址（也叫VTS）
VMAX_ADDR_H	0x	
EXP_ADDR_L	0x	曝光时间控制寄存器的地址
EXP_ADDR_H	0x	
AGAIN_ADDR_L	0x	模拟增益寄存器地址
AGAIN_ADDR_H	0x	
DGAIN_ADDR_L	0x	数字增益寄存器地址
DGAIN_ADDR_H	0x	
WIDTH		Sensor出图宽高信息
HEIGHT		
VMAX_VAL		帧率控制初始值。

vmax_val的计算方式：例如某sensor控制帧率的vmax寄存器为0x3021 0x3022，其初始化序列中设置的初始值为：

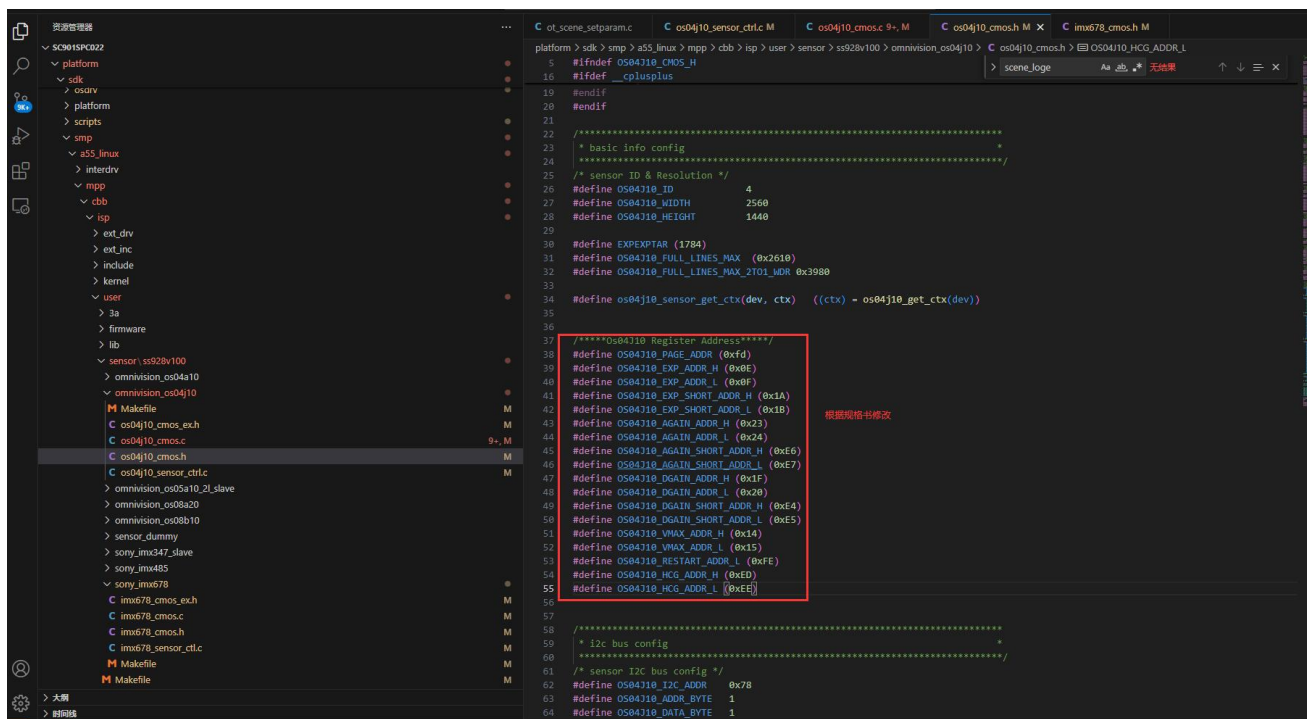
0x3021: 0x06

0x3022: 0x22

则vmax的值为0x0622 = 1570

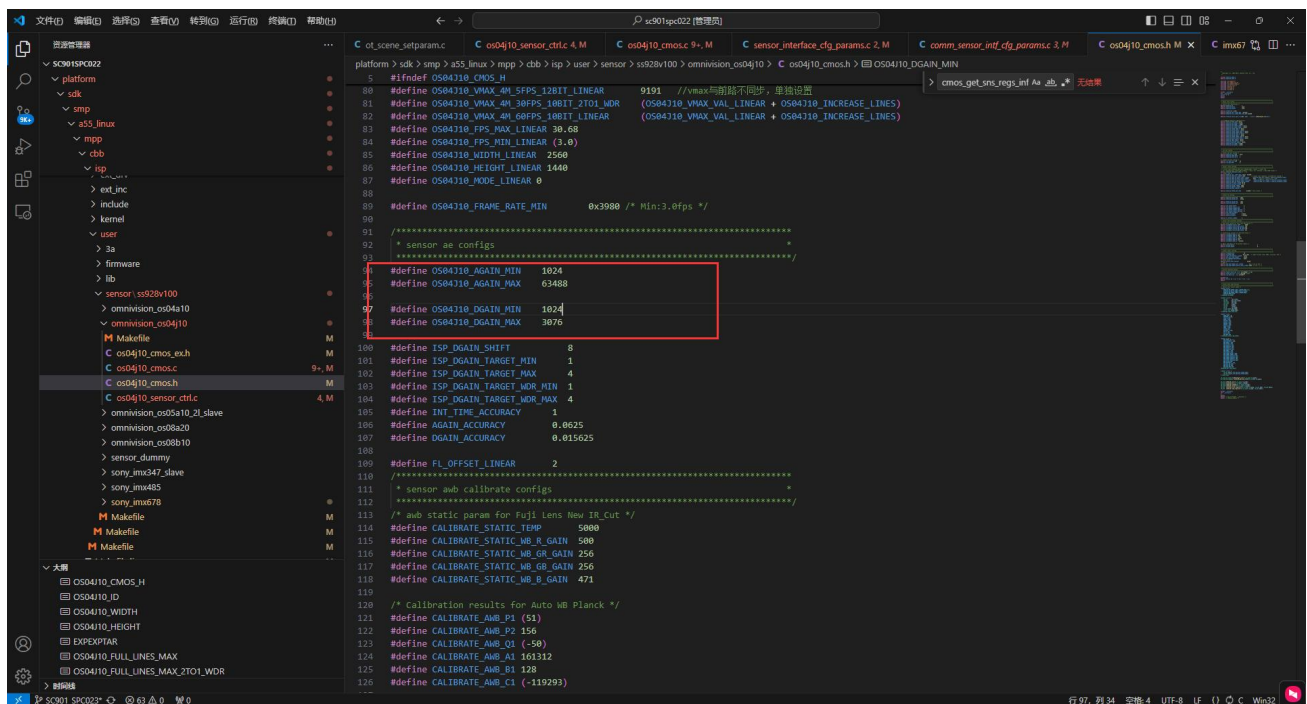
4.1

修改曝光（exposure）、增益（Again、Dgain）、帧率控制（Vmax）寄存器的地址



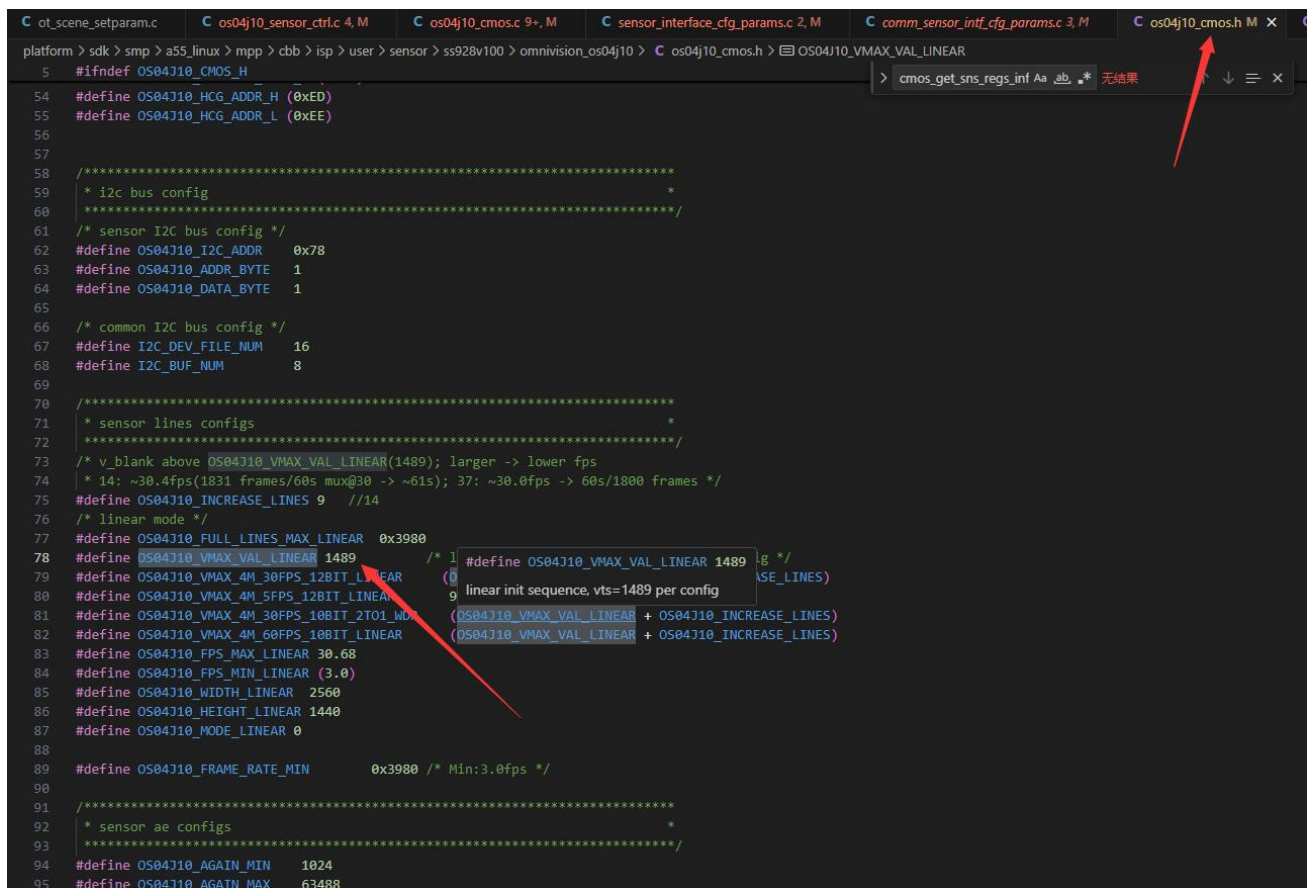
4.2 修改 AE 相关参数

检查最大模拟增益，最小模拟增益，最大数字增益，最小数字增益



4.3 修改 vmax 的值

通常会跟序列申请的序列一起给过来



```
platform > sdk > smp > a55_linux > mpp > cbb > isp > user > sensor > ss928v100 > omnivision_os04j10 > C os04j10_cmos.h > OS04J10_VMAX_VAL_LINEAR
5 #ifndef OS04J10_CMOS_H
54 #define OS04J10_HCG_ADDR_H (0xED)
55 #define OS04J10_HCG_ADDR_L (0xEE)
56
57
58 /******
59 * i2c bus config
60 *****/
61 /* sensor I2C bus config */
62 #define OS04J10_I2C_ADDR 0x78
63 #define OS04J10_ADDR_BYTE 1
64 #define OS04J10_DATA_BYTE 1
65
66 /* common I2C bus config */
67 #define I2C_DEV_FILE_NUM 16
68 #define I2C_BUF_NUM 8
69
70 /******
71 * sensor lines configs
72 *****/
73 /* v_blank above OS04J10_VMAX_VAL_LINEAR(1489); larger -> lower fps
74 * 14: ~30.4fps(1831 frames/60s mux@30 -> ~61s); 37: ~30.0fps -> 60s/1800 frames */
75 #define OS04J10_INCREASE_LINES 9 //14
76 /* linear mode */
77 #define OS04J10_FULL_LINES_MAX_LINEAR 0x3980
78 #define OS04J10_VMAX_VAL_LINEAR 1489 /* 1 #define OS04J10_VMAX_VAL_LINEAR 1489 18 */
79 #define OS04J10_VMAX_4M_30FPS_12BIT_LINEAR (0) /* 0 #define OS04J10_VMAX_4M_30FPS_12BIT_LINEAR 0 */
80 #define OS04J10_VMAX_4M_5FPS_12BIT_LINEAR (0) /* 0 #define OS04J10_VMAX_4M_5FPS_12BIT_LINEAR 0 */
81 #define OS04J10_VMAX_4M_30FPS_10BIT_2T01_LINEAR (OS04J10_VMAX_VAL_LINEAR + OS04J10_INCREASE_LINES)
82 #define OS04J10_VMAX_4M_60FPS_10BIT_LINEAR (OS04J10_VMAX_VAL_LINEAR + OS04J10_INCREASE_LINES)
83 #define OS04J10_FPS_MAX_LINEAR 30.68
84 #define OS04J10_FPS_MIN_LINEAR (3.0)
85 #define OS04J10_WIDTH_LINEAR 2560
86 #define OS04J10_HEIGHT_LINEAR 1440
87 #define OS04J10_MODE_LINEAR 0
88
89 #define OS04J10_FRAME_RATE_MIN 0x3980 /* Min:3.0fps */
90
91 /******
92 * sensor ae configs
93 *****/
94 #define OS04J10_AGAIN_MIN 1024
95 #define OS04J10_AGAIN_MAX 63488
```

4.4 模拟增益、数字增益

检查这两个表格是否正确，检查计算的 gain 是否正确。again_lin, again_db, dgain_lin, dgain_db 既是输入参数，也是输出参数。当作输入时，去匹配 gain table 中的值，修正或换算后，当作输出，传入 cmos_gains_update(ot_vi_pipe vi_pipe, td_u32 again, td_u32 dgain)函数设置寄存器的值。由于 gain 的计算涉及对数运算，比较消耗算力，因此使用查表法。Gain table 可以找 sensor 原厂获取。

```
C ot_scene_setparam.c C os04j10_sensor_ctrlc 4, M C os04j10_cmos.c 9+, M X C sensor_interface_cfg_params.c 2, M C comm_sensor_intf_cfg_params.c 3, M C os04j10_cmos.h M
platform > sdk > smp > a55_linux > mpp > cbb > isp > user > sensor > ss928v100 > omnivision_os04j10 > C os04j10_cmos.c
76 #define OS04J10_DGAIN_TAB_ROW 31
77
78 /*****
79 Again & Dgain table for TABLE Mode
80 厂商 Gain Table: again 步长 0x10(16), dgain 步长 0x40(64)
81 again 寄存器值 = g_a_gain_table[idx] / 0x40, 范围 0x010~0x3E0 (1x~62x)
82 dgain 寄存器值 = g_d_gain_table[idx] / 0x10, 范围 0x040~0x7FF (1x~32x)
83 *****/
84 /* dgain: 每档 para*64, 寄存器值 = table[idx]/0x10 */
85 static td_u32 g_d_gain_table[OS04J10_DGAIN_TAB_ROW + 2] = {
86     /* para 1~32, 寄存器值 0x040~0x7FF */
87     1024, 2048, 3072, 4096, 5120, 6144, 7168, 8192,
88     9216, 10240, 11264, 12288, 13312, 14336, 15360, 16384,
89     17408, 18432, 19456, 20480, 21504, 22528, 23552, 24576,
90     25600, 26624, 27648, 28672, 29696, 30720, 31744, 32752,
91     32768
92 };
93
94 /* again: 每档 para*16, 寄存器值 = table[idx]/0x40 */
95 static td_u32 g_a_gain_table[OS04J10_AGAIN_TAB_ROW + 2] = {
96     /* para 1~62, 寄存器值 0x010~0x3E0 */
97     1024, 2048, 3072, 4096, 5120, 6144, 7168, 8192,
98     9216, 10240, 11264, 12288, 13312, 14336, 15360, 16384,
99     17408, 18432, 19456, 20480, 21504, 22528, 23552, 24576,
100     25600, 26624, 27648, 28672, 29696, 30720, 31744, 32768,
101     33792, 34816, 35840, 36864, 37888, 38912, 39936, 40960,
102     41984, 43008, 44032, 45056, 46080, 47104, 48128, 49152,
103     50176, 51200, 52224, 53248, 54272, 55296, 56320, 57344,
104     58368, 59392, 60416, 61440, 62464, 63488
105 };
106
107 /*****
108 * common functions
109 *****/
110 ot_isp_sns_commbus *os04j10_get_bus_info(ot_vi_pipe vi_pipe)
111 {
112     if (vi_pipe < 0 || vi_pipe >= OT_ISP_MAX_PIPE_NUM) {
113         return TD_NULL;
114     }
115     return &g_bus_info[vi_pipe];
116 }
117
118 ot_isp_sns_state *os04j10_get_ctx(ot_vi_pipe vi_pipe)
119 {
120     if (vi_pipe < 0 || vi_pipe >= OT_ISP_MAX_PIPE_NUM) {
121         return TD_NULL;
122     }
123 }
```

OS04J10 Analog Gain table(1.0x~62x)				OS04J10 Digit Gain table(1.0x~32x)				exposure		
long gain	寄存器		Again (0x) STEP: 1/16	long gain	寄存器		Dgain (0x) STEP: 1/64		register	value
	P10x03	P10x04			P10x1F	P10x20				
short gain	P10x66	P10x67		short gain	P10x64	P10x65				
	0x00	0x10	1	0x00	0x40	1		long	P1:0x0e	05
	0x00	0x20	2	0x00	0x80	2			P1:0x0f	a0
	0x00	0x30	3	0x00	0xc0	3		short	P1:0x1e	00
	0x00	0x40	4	0x01	0x00	4			P1:0x1f	08
	0x00	0x50	5	0x01	0x40	5		Min Digital Gain = 0x040(1xgain)Max Digital Gain = 0x7ff(32xgain) Min Analog Gain = 0x010(1xgain)Max Analog Gain = 0x3e0(62xgain) Min Exposure = 2line Max Exposure = VTS - delta_reg HDR Mode: long Exposure + short Exposure = new Exposure		
	0x00	0x60	6	0x01	0x80	6				
	0x00	0x70	7	0x01	0xc0	7				
	0x00	0x80	8	0x02	0x00	8				
	0x00	0x90	9	0x02	0x40	9				
	0x00	0xa0	10	0x02	0x80	10				
	0x00	0xb0	11	0x02	0xc0	11				
	0x00	0xc0	12	0x03	0x00	12				
	0x00	0xd0	13	0x03	0x40	13				
	0x00	0xe0	14	0x03	0x80	14				
	0x00	0xf0	15	0x03	0xc0	15				
	0x01	0x00	16	0x04	0x00	16				
	0x01	0x10	17	0x04	0x40	17				
	0x01	0x20	18	0x04	0x80	18				
	0x01	0x30	19	0x04	0xc0	19				
	0x01	0x40	20	0x05	0x00	20				
	0x01	0x50	21	0x05	0x40	21				
	0x01	0x60	22	0x05	0x80	22				
	0x01	0x70	23	0x05	0xc0	23				
	0x01	0x80	24	0x06	0x00	24				
	0x01	0x90	25	0x06	0x40	25				
	0x01	0xa0	26	0x06	0x80	26				
	0x01	0xb0	27	0x06	0xc0	27				
	0x01	0xc0	28	0x07	0x00	28				
	0x01	0xd0	29	0x07	0x40	29				
	0x01	0xe0	30	0x07	0x80	30				
	0x01	0xf0	31	0x07	0xc0	31				
	0x02	0x00	32	0x07	0xff	32				
	0x02	0x10	33							
	0x02	0x20	34							
	0x02	0x30	35							
	0x02	0x40	36							
	0x02	0x50	37							
	0x02	0x60	38							
	0x02	0x70	39							
	0x02	0x80	40							
	0x02	0x90	41							
	0x02	0xa0	42							
	0x02	0xb0	43							
	0x02	0xc0	44							
	0x02	0xd0	45							
	0x02	0xe0	46							
	0x02	0xf0	47							
	0x03	0x00	48							
	0x03	0x10	49							
	0x03	0x20	50							
	0x03	0x30	51							
	0x03	0x40	52							
	0x03	0x50	53							
	0x03	0x60	54							
	0x03	0x70	55							

4.5 HCG 和 LCG 的切换

如何确定 HCG 和 LCG 切换倍数，需要问 sensor 原厂如下的寄存器读取，这个值一定要准确，否则切换时会闪烁

DCG

➤ OS04J10 support dynamic conversion gain switch, and we can read CG ratio from OTP. Follow is the CG ratio read procedure:

1. Load follow register:

```
78 fd 00 ;  
78 57 10 ;  
sl 1 1 ;  
sl 1 1 ;  
sl 1 1 ;
```

```
78 fd 09 ;  
78 9a 00 ;  
78 5d 00 ;  
78 58 00 ;  
78 89 00 ;  
78 8a 00 ;  
78 8b 17 ;  
78 84 40 ;  
78 81 01 ; delay 7us
```

2. Read page C:0x00, then the CG ratio=(the value/0x20), the CG ratio range is about 4.4~5.

➤ Follow is the HCG and LCG switch register:

```
78 fd 01 ; page1  
78 ed ; [0] linear mode or staggerHDR mode long frame 1:HCG, 0:LCG  
78 ee ; [0] staggerHDR mode short frame 1:HCG, 0:LCG  
78 fe 02 must enable after fe=02  
DCG mode don't support HCG/CG switch
```

写个脚本

```
#!/bin/sh
```

```
# OS04J10 DCG CG Ratio 读取脚本
```

```
# 平台: HI3519AV200
```

```
# I2C bus: 2, I2C addr: 0x78, reg_width: 1, data_width: 1
```

```
# 用法: i2c_write <i2c_num> <dev_addr> <reg_addr> <value> <reg_width>  
<data_width>
```

```
# i2c_read <i2c_num> <dev_addr> <reg_addr> <reg_addr_end> <reg_width>  
<data_width>
```

```
I2C=0x2
```

```
DEV=0x78
```

```
echo "===== OS04J10 DCG CG Ratio 读取 ====="
```

---- 步骤 1: 写入 OTP 读取准备寄存器 ----

echo "[步骤 1] 写入 OTP 读取准备寄存器..."

i2c_write \$I2C \$DEV 0xfd 0x00 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x57 0x10 > /dev/null 2>&1

sleep 1

sleep 1

sleep 1

i2c_write \$I2C \$DEV 0xfd 0x09 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x9a 0x00 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x9d 0x00 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x88 0x00 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x89 0x00 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x8a 0x00 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x8b 0x17 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x84 0x40 > /dev/null 2>&1

i2c_write \$I2C \$DEV 0x81 0x01 > /dev/null 2>&1

usleep 72 > /dev/null || sleep 1

echo "[步骤 1] 完成"

```
# ---- 步骤 2: 读取 page 0x0C reg 0x00 ----
```

```
echo "[步骤 2] 读取 CG ratio..."
```

```
i2c_write $I2C $DEV 0xfd 0x0c > /dev/null 2>&1
```

```
# 只读 0x00 寄存器, 提取值
```

```
RAW=$(i2c_read $I2C $DEV 0x00 0x00 2>/dev/null)
```

```
# 从输出中提取 0xNN 格式的值
```

```
HEX_VAL=$(echo "$RAW" | grep "0x0:" | sed 's/.*0x0: //' | tr -d ' \r\n')
```

```
if [ -z "$HEX_VAL" ]; then
```

```
    echo "[错误] 读取失败, 无法获取寄存器值"
```

```
    i2c_write $I2C $DEV 0xfd 0x00 > /dev/null 2>&1
```

```
    exit 1
```

```
fi
```

```
# 转换为十进制
```

```
DEC_VAL=$(printf "%d" "$HEX_VAL")
```

```
# 计算 CG ratio = value / 32, 保留 6 位小数
```

```
# busybox ash 没有浮点, 用整数模拟: ratio = val * 1000000 / 32
```

```
RATIO_X1M=$(( DEC_VAL * 1000000 / 32 ))
```

```
RATIO_INT=$(( RATIO_X1M / 1000000 ))
```

```
RATIO_FRAC=$(( RATIO_X1M % 1000000 ))
```

```
# 补零到 6 位
```

```
RATIO_FRAC_STR=$(printf "%06d" "$RATIO_FRAC")
```

```
# ---- 恢复 page ----
```

```
i2c_write $I2C $DEV 0xfd 0x00 > /dev/null 2>&1
```

```
echo ""
```

```
echo "==== 结果 ====="
```

```
echo "Page 0x0C Reg 0x00 = $HEX_VAL ($DEC_VAL)"
```

```
echo "CG ratio = $DEC_VAL / 32 = ${RATIO_INT}.${RATIO_FRAC_STR}"
```

```
echo ""
```

```
if [ $RATIO_X1M -ge 4400000 ] && [ $RATIO_X1M -le 5000000 ]; then
```

```
    echo "状态: 正常 (范围 4.40 ~ 5.00)"
```

```
else
```

```
    echo "状态: 异常! (期望范围 4.40 ~ 5.00)"
```

```
fi
```

```
echo "=====
```

运行读取结果如下：

```
/app/bin #  
/app/bin #  
/app/bin # ./read_os04j10_cg_ratio.sh  
===== OS04J10 DCG CG Ratio 读取 =====  
[步骤1] 写入 OTP 读取准备寄存器...  
[步骤1] 完成  
[步骤2] 读取 CG ratio...  
  
===== 结果 =====  
Page 0x0C Reg 0x00 = 0x95 (149)  
CG ratio = 149 / 32 = 4.656250  
  
状态: 正常 (范围 4.40 ~ 5.00)  
=====
```

```
C ot_scene_setparam.c  C os04j10_sensor_ctrl.c 4.M  C os04j10_cmos.c 9+.M X  C sensor_interface_cfg_params.c 2.M  C comm_sensor_intf_cfg_params.c 3.M  C os04j10_cmos.h M  
platform > sdk > smp > a55_linux > mpp > cbb > isp > sensor > ss928v100 > omnivision_os04j10 > C os04j10_cmos.c > cmos_gains_update(ot_vi_pipe, td_u32, td_u32)  
624  ot_isp_agc_int_time_range *int_time, td_u32 *lf_max_int_time)  
625  {  
626  // ot_isp_agc_int_time_range *int_time = 0;  
627  return;  
628  }  
629  
630 static td_void cmos_gains_update(ot_vi_pipe vi_pipe, td_u32 pu32_again, td_u32 pu32_dgain)  
631 {  
632  td_u32 again, dgain;  
633  static td_bool first_gain[OT_ISP_MAX_PIPE_NUM] = { [0 ... (OT_ISP_MAX_PIPE_NUM - 1)] = 1 };  
634  
635  ot_isp_sns_state *sns_state = TD_NULL;  
636  //sns_check_pipe_void_return(vi_pipe);  
637  
638  sns_state = os04j10_get_ctx(vi_pipe);  
639  sensor_check_pointer_void_return(sns_state);  
640  
641  again = (g_a_gain_table[pu32_again] / 0x40);  
642  dgain = (g_d_gain_table[pu32_dgain] / 0x10);  
643  
644  if (sns_state->wdr_mode == OT_WDR_MODE_NONE) {  
645  /* HCG 切换逻辑: again_reg >= 0x4B(4.66x) 时打开 HCG, HCG 增益 4.66 倍 (OTP CG ratio: 0x95/0x20) */  
646  
647  if (again >= 0x4B) {  
648  // printf("os04j10 HCG ON: idx=%u, again_before=0x%03X, again_after=0x%03X, dgain=0x%03X\n",  
649  // pu32_again, again, (td_u32)((td_float)again / 4.66 + 0.5), dgain);  
650  
651  again = (td_u32)((td_float)again / 4.66 + 0.5);  
652  sns_state->regs_info[0].i2c_data[HCG_H_IDX].data = 0x01;  
653  sns_state->regs_info[0].i2c_data[HCG_L_IDX].data = 0x01;  
654  } else {  
655  // printf("os04j10 LCG: idx=%u, again=0x%03X, dgain=0x%03X\n",  
656  // pu32_again, again, dgain);  
657  
658  sns_state->regs_info[0].i2c_data[HCG_H_IDX].data = 0x00;  
659  sns_state->regs_info[0].i2c_data[HCG_L_IDX].data = 0x00;  
660  }  
661  
662  sns_state->regs_info[1].i2c_data[VALUE_IDX].data = 0x00; /* index 2 (0x02cd) */  
663  sns_state->regs_info[0].i2c_data[PAGE_IDX].data = 0x01; /* index 3, group hold */  
664  
665  sns_state->regs_info[0].i2c_data[AGAIN_L_IDX].data = low_8bits(again); /* index 4, index 1 (0x0614) */  
666  sns_state->regs_info[0].i2c_data[AGAIN_H_IDX].data = high_8bits(again);  
667  
668  sns_state->regs_info[0].i2c_data[DGAIN_L_IDX].data = low_8bits(dgain);  
669  sns_state->regs_info[0].i2c_data[DGAIN_H_IDX].data = high_8bits(dgain);  
670  
671  sns_state->regs_info[0].i2c_data[VALUE_IDX].data = 0x01; /* index 5, index 2 (0x0615) */  
672  }
```

第5章 等曝光量测试

参考《pq 环境搭建.docx》文档搭建 PQ 环境，然后根据连接 PQ 工具进行等曝光量测试，

验证曝光和增益是同时进行的，基本原理如下：

(1) 曝光时间为 10000，增益为 1 倍

(2) 曝光时间为 2500，增益为 4 倍

上述两中设定的曝光量是一样的，所有对应的画面的亮度也应该是一样的，做等曝光量测试就是再 pqtools 设置这两种曝光量的切换，要求画面的亮度不能有变化，切换时画面也不能有闪烁。

步骤一：将下面的图片中的自动参数，全部设为手动；

步骤二：点击 Float 之后，gain 的参数变为倍数设定，设置 1 为一倍，设置 2，为 2 倍，设置 4 为 4 倍增益。

步骤三：点击扩展按钮，出现 to set1 和 to set2。我们可以设置曝光时间为 10000，增益为 1 倍，随后点击设定 to set1；设定曝光时间为 2500，增益为 4 倍，随后点击 to set2。此时 load set1 和 load set2 都可以点击了。这时可以切换两种曝光量参数，查看 RTSP 拉流的画面，看亮度是否保持不变，画面不能有闪烁。

PQTools [10.85.1.8]

Frequency: BASIC

HI3516CV610 (1.0.1.0)

Top

Pub Attr

CAC

BLC

WB Attr

WB Info

CCM

CA

Saturation

AE Route

Exposure Attr

Exposure Info

DRC

Gamma

LDCI

Dehaze

Defect Pixel

BNR

Demosaic

3DNR Attr

3DNR X

Sharpen

Gop Mode

<Select an Extra Component>

All Read

All Write

Read Page

Write Page

Auto Write

Read

Write

Default

isp_exposure_attr

Read

Write

Default

bypass	☐ Enable ☒ Disable
op_type	OP_MODE_MANUAL
ae_run_interval	1
hist_stat_adjust	☒ Enable ☐ Disable
ae_route_ex_valid	☐ Enable ☒ Disable
prior_frame	ISP_LONG_FRAME
ae_gain_sep_cfg	☐ Enable ☒ Disable
advance_ae	☐ Enable ☒ Disable

isp_me_attr

Read

Write

Float

Default

To Set1

To Set2

Load Set1

Load Set2

exp_time_op_type	OP_MODE_MANUAL
a_gain_op_type	OP_MODE_MANUAL
d_gain_op_type	OP_MODE_MANUAL
ispd_gain_op_type	OP_MODE_MANUAL
exp_time	16384
a_gain	1.00
d_gain	1.00
ispd_gain	1.00

isp_ae_attr

Read

Write

Float

Default

exp_time_range.max	1456333
exp_time_range.min	44
a_gain_range.max	32768
a_gain_range.min	1024
d_gain_range.max	16128
d_gain_range.min	1024
ispd_gain_range.max	32768
ispd_gain_range.min	1024
sys_gain_range.max	16515072
sys_gain_range.min	1024
gain_threshold	16515072
speed	64
black_speed_bias	144
tolerance	2
compensation	64
ev_bias	1024
ae_strategy_mode	ISP_AE_EXP_HIGHLIGHT_PRIOR
hist_ratio_slope	128
max_hist_offset	16
ae_mode	ISP_AE_MODE_FIX_FRAME_RATE
antiflicker.enable	☐ Enable ☒ Disable
antiflicker.frequency	50
antiflicker.mode	ISP_ANTIFlicker_NORMAL_MODE
subflicker.enable	☐ Enable ☒ Disable

Log Viewer

System

Clear All

☒ Scroll to bottom when the logs refreshed

☐ Save log to file

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] [0x011103][11] Create connection...

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] [0x011103][11] Sending request...

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] Total Packet length: 160

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] Operation length: 32

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] [0xFF011103][11] Receiving response...

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] [0xFF011103][11] Receiving response...

[2025/01/14 19:36:23][Network 10.85.1.8(4321)] [Exposure Attr/isp_me_attr] [INFO] [0xFF011103][11] Communication finished.

Views : PQ Table Log Viewer

Ready

PQTools [disconnect]

Frequency: BASIC

Exposure Attr

isp_exposure_attr

Read Write Default

bypass ☐ Enable ☒ Disable

op_type **OP_MODE_MANUAL**

ae_run_interval 1

hist_stat_adjust ☐ Enable ☒ Disable

ae_route_ex_valid ☐ Enable ☒ Disable

prior_frame ISP_LONG_FRAME

ae_gain_sep_cf ☐ Enable ☒ Disable

advance_ae ☐ Enable ☒ Disable

isp_ae_attr

Read Write + Float Default

exp_time_range.max 0

exp_time_range.min 0

a_gain_range.max 1024

a_gain_range.min 1024

d_gain_range.max 1024

d_gain_range.min 1024

ispd_gain_range.max 1024

ispd_gain_range.min 1024

sys_gain_range.max 1024

sys_gain_range.min 1024

gain_threshold 1024

isp_me_attr

Read Write + Float Default

To Set1 To Set2 Load Set1 Load Set2

exp_time_op_type **OP_MODE_MANUAL**

a_gain_op_type **OP_MODE_MANUAL**

d_gain_op_type **OP_MODE_MANUAL**

ispd_gain_op_type **OP_MODE_MANUAL**

exp_time 0

a_gain 1.00

d_gain 1.00

isp_d_gain 1.00

black_speed_bias 0

tolerance 0

compensation 0

ev_bias 0

ae_strategy_mode ISP_AE_EXP_HIGHLIGHT_PRIOR

hist_ratio_slope 0

max_hist_offset 0

ae_mode ISP_AE_MODE_SLOW_SHUTTER

antiflicker.enable ☐ Enable ☒ Disable

antiflicker.frequency 0

antiflicker.mode ISP_ANTIFLICKER_NORMAL_MODE

1、设置五个地方为手动模式

2、点击下拉，可以设set1 set2两个参数，

3、点击float抓化为比例设置模式

Expand 'Advance Settings' panel.

Views : PQ Table Log Viewer

Ready

-- 文档结束 --